

Calvin Trees – Interactive Campus Arboretum

Vision Statement

Our goal is to continue development and complete production of Professor Norman's existing Calvin Trees app. The vision is to engage various demographics (ranging from college students to campus visitors of all ages) with our campus, specifically the natural side of Calvin University. We will accomplish this by polishing existing UI elements, finishing implementation of partially completed features, and introduce brand new features altogether. Key ethical and design norms for our project include cultural appropriateness and aesthetic norms. It is imperative that we develop something that can foster interest in our campus tree ecosystem without disturbing the ecosystem or cause distraction and, consequently, detract from the entire purpose of engaging with nature. Moreover, the app and new functionalities need to be intuitive and easy to understand to make the core purpose of exploring Calvin University's trees enjoyable.

Normative and Ethical Consideration

Social

A potential challenge we could have with this norm is that our project could take away from engagement with Calvin University's natural environment. Creating a mobile application that distracts from truly appreciating our campus' surroundings. We can combat this by implementing simple UI and focusing on guiding without too many pop ups and visual interruptions.

Aesthetics

We need to ensure that our application is aesthetically pleasing to engage with, and intuitive enough to allow seamless usage and maximize engagement with our campus environment. Our team can accomplish this by adhering to key UI design principles such as maintaining consistency, utilizing contrast efficiently, making the application's features accessible and simple, and conduct user tests to ensure we are not biased.

Cultural Appropriateness

Our app should encourage meaningful interaction with the environment while ensuring that Calvin University's ecosystem remains undisturbed. We can achieve this by incorporating guidance and gentle reminders for users. These could include informative messages that promote responsible behavior, such as properly disposing of trash and respecting natural spaces.

Background

Traditional arboretum and environmental engagement platforms are often designed for large botanical gardens, national parks, or funded institutions with significant budgets and staff. Examples include iNaturalist which provides powerful tools for mapping, species tracking, and ecological data collection. While robust in functionality, these are typically built for broader scientific or large-scale public engagement and requires substantial ongoing maintenance and funding.

For smaller campus-based engagement like with Calvin University, these solutions may be too complex, expensive, and/or misaligned with this institution's goals. The primary objective is to foster student engagement, environmental awareness, and appreciation of the campus' natural ecosystem. A generalized platform may not accomplish this. Our application is highly focused and accessible, especially for mobile users, because of our progressive web application implementation. Rather than being a scientific database, Calvin Trees focuses on proximity-based notifications, randomly generated waking tours, and simple interactive guidance. Our implementation ensures minimal ongoing maintenance costs and requirements.

Development Process

We are storing source code on GitHub, meeting with our team and professor weekly (Thursday), and commit 6 hours weekly as a minimum.

- GitHub
 - o Production Branch → currently gets deployed via Firebase Hosting
 - o Dev Branch → individuals branch from here and use PRs for merging code
 - o Quality Checks → implement linter and code reviews for coding conventions
- Planning and Organization
 - o Notion → organize schedules and broader backlog tasks
 - o GitHub Projects → organize tasks with greater specificity, delegate tasks

Contributions

- Sam
 - o Fixed numerous bugs including a duplicate search results bug, fixed the marker and location implementation, and assisted in enhanced error handling

- Peter
 - Investigated initial implementation, fixed layout and modals for individual trees, updated our tree data
- Alim
 - Built function CRUD for tree data and admin capability, full testing coverage, and implementation of two tour mode

Success Criteria

- Development of fully functional PWA that reliably delivers all core features including:
 - Interactive mapping
 - Proximity-based notifications
 - Information tree components
 - Guided walking tours
- Successful deployment:
 - Public release
 - Clear communication (e.g. via flyers, website posting) that app is ready for students, faculty, and campus visitors